chapter-3

RAJIB
che-50

2021, 2018, 2019(H)

(H) Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic & an anharmonic oscillators. show allowed transitions & the expected spectrum?

⇒ schematic diagram for harmonic oscillators:

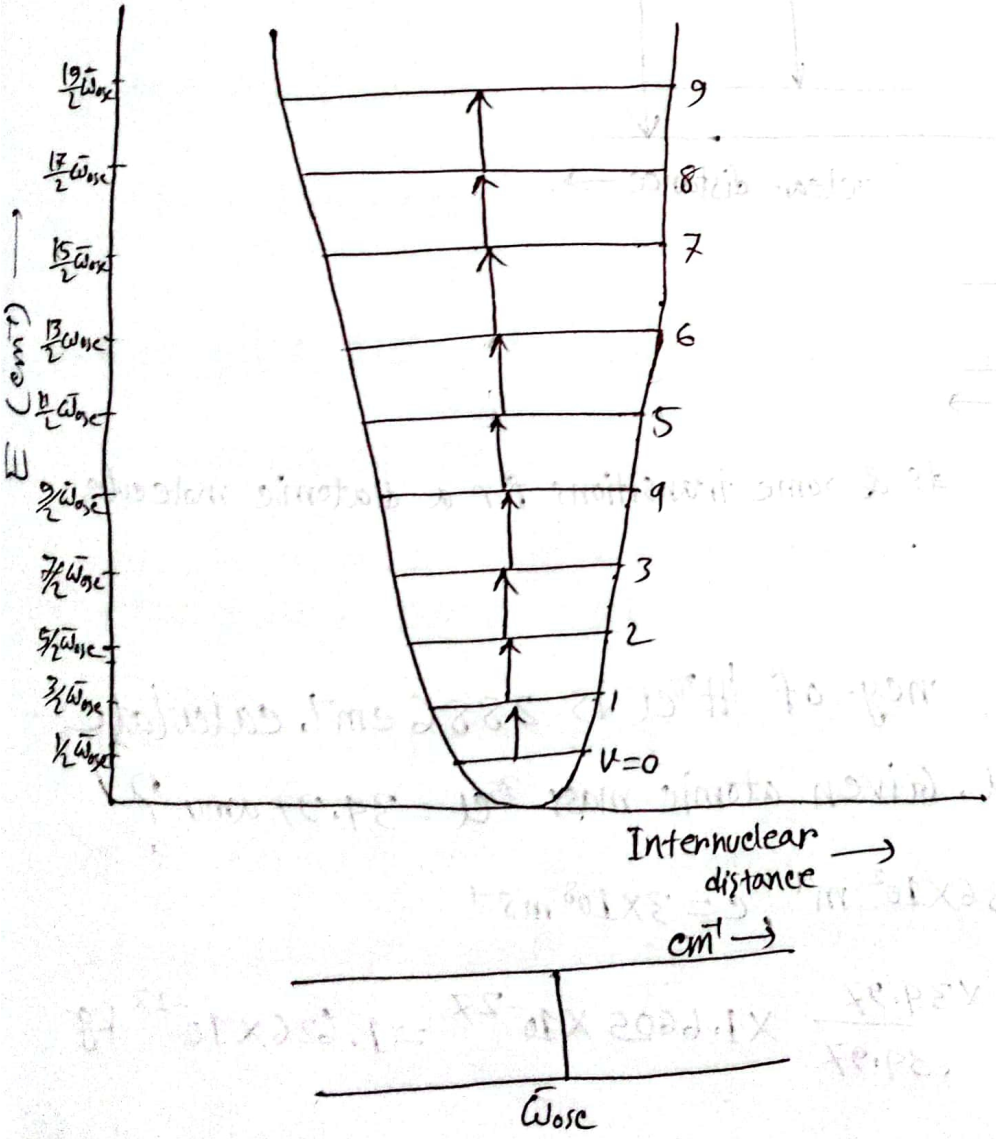

Fig: The vibrational energy levels & allowed transitions btwn them for a diatomic molecule undergoing simple harmonic motion.

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Schematic diagram for anharmonic oscillator:

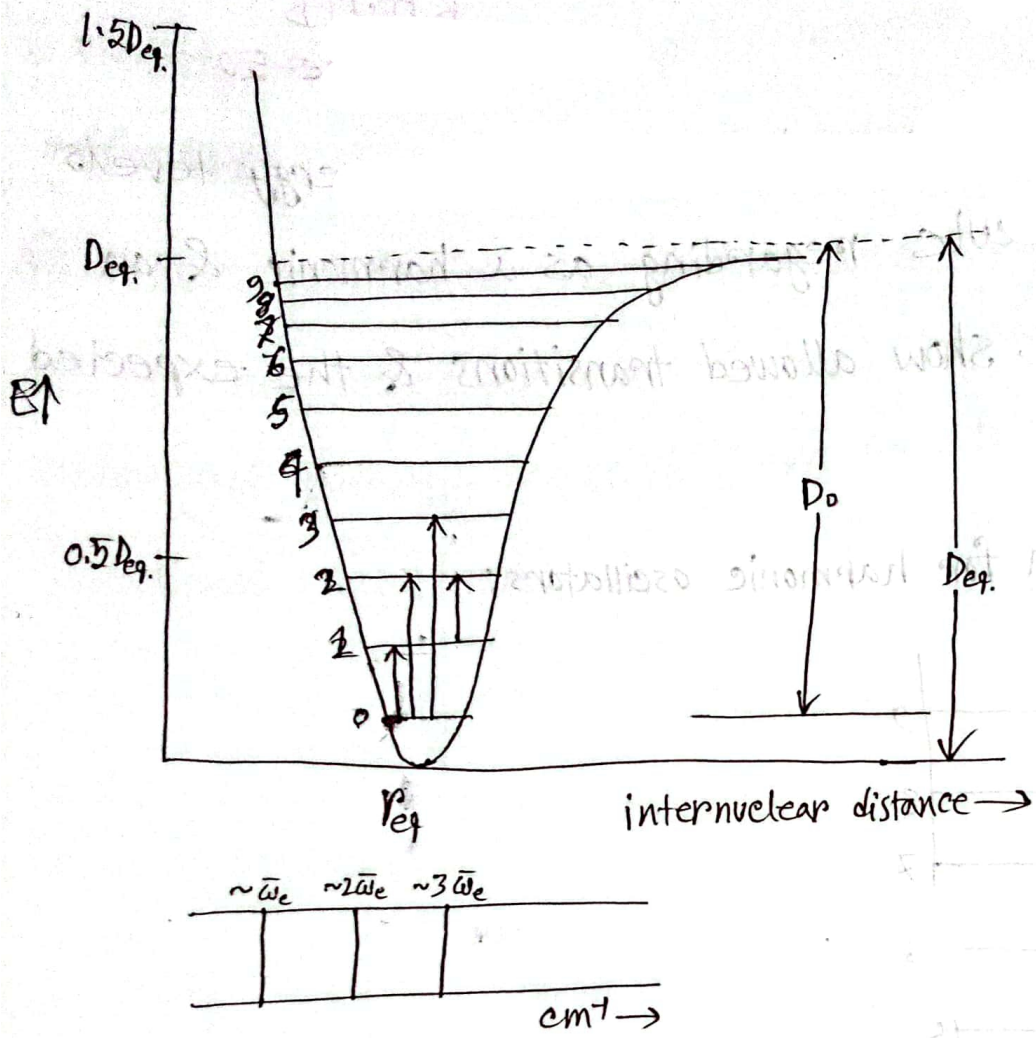


Fig: The vibrational energy levels & some transitions for a diatomic molecule undergoing anharmonic oscillation.

2016, 2018(t)

⊛ The fundamental frequency of $^1H^{35}Cl$ is 2886 cm^{-1}. calculate force constant of $^1H^{35}Cl$. Given atomic mass ^{35}Cl = 34.97 amu ?

$\Rightarrow \bar{\omega}_e = 2886\ cm^{-1} = 2886 \times 10^2\ m^{-1}$, $c = 3 \times 10^8\ ms^{-1}$

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{1.0078 \times 34.97}{1.0078 * 34.97} \times 1.6605 \times 10^{-27} = 1.626 \times 10^{-27}\ kg$$

$\therefore$ Force constant, $K = 4\pi^2 \bar{\omega}_e^2 c^2 \mu = 481.219\ Nm^{-1}$ Ans.

2017, 2020, 2021(t)

(*) The equilibrium oscillation frequency for NO is 1904 cm^{-1}. calculate force constant of the molecule?

$\Rightarrow \bar{\omega}_e = 1904 \times 10^2 m^{-1}$, $c = 3 \times 10^8 ms^{-1}$,

$$\mu = \left(\frac{14 \times 16}{14+16}\right) \times 1.6605 \times 10^{-27} = 1.24 \times 10^{-26} kg$$

$\therefore K = 4\pi^2 \bar{\omega}_e^2 c^2 \mu = 1597.19 \ Nm^{-1}$ Ans.

2018, 2021, 2019(t)

(*) the fundamental band of CO in inert CCl_4 is 2143 cm^{-1}. calculate K for CO?

$\Rightarrow \bar{\omega}_e = 2143 \times 10^2 m^{-1}$, $\mu = \left(\frac{12 \times 16}{12+16}\right) \times 1.6605 \times 10^{-27} = 1.14 \times 10^{-27} kg$

$\therefore K = 4 \times (3.1416)^2 \times (214300)^2 \times (3 \times 10^8)^2 \times 1.14 \times 10^{-27}$

$= 1860.16 \ Nm^{-1}$

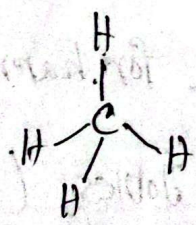
2017, 2020, 2022, 2021(t), 2018(t)

(*) How many normal modes of vibration are possible of : $CH \equiv CH$ CH_4, HBr, O_2, OCS (linear), SO_2 (bent), $CH_2 = CH_2$, NH_3, C_6H_6, H_2O, BCl_3, CH_3I ?

$\Rightarrow$ (i) for $CH \equiv CH$, $\rightarrow$ $H - C \equiv C - H$

It is linear molecule. So normal mode of vibration $= 3N - 5$
$= 3 \times 4 - 5$
$= 7$

(ii) For CH_4,



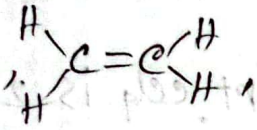
It is non linear. So mode of vibration $= 3N - 6 = 3 \times 5 - 6 = 9$

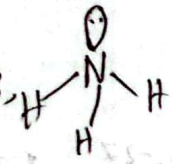
(III) For HBr, H–Br, it is linear. So, mode = 3N−5
= 3×2−5
= 1

(IV) For O_2, O=O, it is linear, so mode = 3N−5 = 3×2−5 = 1.

(V) For OCS, O=C=S, it is linear. So mode = 3×3−5 = 4

(VI) For SO_2, , it is non linear. So mode = 3×3−6 = 3

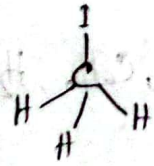
(VII) For $CH_2=CH_2$,  , it is non linear, so mode = 3×6−6
= 12

(VIII) For NH_3,  , it is non linear. So mode = 3×4−6 = 6

(IX) For C_6H_6, , it is non linear. So mode = 3×12−6 = 30

(X) For H_2O, , it is non linear. So mode = 3×3−6 = 3

(XI) For BCl_3, , it is non linear. So mode = 3×4−6 = 6

(XII) For CH_3I,  , it is non linear, so mode = 3×5−6 = 9

(X) 2020, 2017, 2015, 2021(t) write eqn ~~of~~ vibrational energy of a simple harmonic oscillator? From this eqn show diatomic molecule can never have zero vibrational energy? selection rule 2019(t) for harmonic oscillator?

⇒ vibrational energy, $E_v = \left(v + \frac{1}{2}\right) h\omega_{osc}$ Joules $(v = 0, 1, 2, \ldots)$

where, V = vibrational quantum number. converting to the spectroscopic units, cm^{-1}, we have:

$$\varepsilon_v = \frac{E_v}{hc} = \left(V + \frac{1}{2}\right)\frac{\omega_{osc.}}{c} = \left(V + \frac{1}{2}\right)\bar{\omega}_{osc.}\ cm^{-1}$$

⇒ we should notice that the lowest vibrational energy, obtained by putting V = 0,

$$\therefore E_0 = \frac{1}{2} h\omega_{osc.} \text{ Joules } (\omega_{osc.} \text{ in Hz})$$

or, $$\varepsilon_0 = \frac{1}{2}\bar{\omega}_{osc.}\ cm^{-1} \quad (\bar{\omega}_{osc.} \text{ in } cm^{-1})$$

So we can say that, the diatomic molecule can never have zero vibrational energy.

⇒ selection rule for harmonic oscillator, $\Delta V = \pm 1$

2016
~~⊗~~ Sketch the Morse curve for anharmonic oscillator? Give selection rule for anharmonic oscillator? Explain fundamental, hot band & overtone transitions with energy expressions?

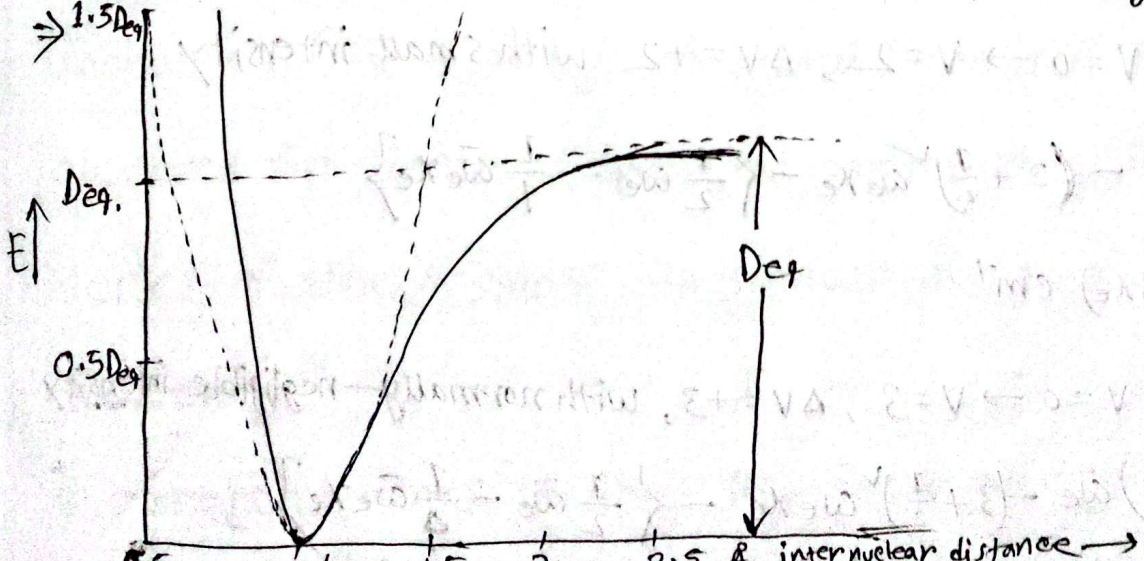


Fig: The Morse curve of diatomic molecule undergoing anharmonic extensions & compressions.

⇒ selection rule for anharmonic oscillator, $\Delta V = \pm 1, \pm 2, \pm 3, \ldots\ldots$

⇒ vibrational energy for anharmonic oscillator,

$$\varepsilon_v = \left(V+\frac{1}{2}\right)\bar{\omega}_e - \left(V+\frac{1}{2}\right)^2 \bar{\omega}_e x_e \ \text{cm}^{-1} \quad (V = 0, 1, 2, \ldots\ldots)$$

where, $\bar{\omega}_e$ = oscillation frequency, x_e = anharmonicity constant

Fundamental absorption: Fundamental vibrational frequencies of a molecule correspond to transition from $V=0$ to $V=1$ with considerable intensity.

$$\Delta\varepsilon = \varepsilon_{V=1} - \varepsilon_{V=0} = \left(1+\frac{1}{2}\right)\bar{\omega}_e - \left(1+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= \bar{\omega}_e (1 - 2x_e) \ \text{cm}^{-1}$$

overtone band: An overtone band is the spectral band that occur in vibrational spectrum of a molecule when the molecule makes a transition from the ground state $V=0$ to the second or 3rd excited state ($V=2$ or $V=3$).

For 1st overtone: $V=0 \rightarrow V=2$, $\Delta V = +2$ with small intensity

$$\Delta\varepsilon = \left(2+\frac{1}{2}\right)\bar{\omega}_e - \left(2+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= 2\bar{\omega}_e (1 - 3x_e) \ \text{cm}^{-1}$$

For 2nd overtone: $V=0 \rightarrow V=3$, $\Delta V = +3$, with normally negligible intensity

$$\Delta\varepsilon = \left(3+\frac{1}{2}\right)\bar{\omega}_e - \left(3+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= 3\bar{\omega}_e (1 - 4x_e) \ \text{cm}^{-1}$$

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Hot bands: The weak absorption arise at slightly lower wave number than the fundamental, such weak absorption is known as hot bands. A high temperature is the preliminary conditions for their appearance. $V=1 \longrightarrow V=2$, $\Delta V=+1$, normally very weak

$$\Delta E = \left(2+\frac{1}{2}\right)\bar{\omega}_e - \left(2+\frac{1}{2}\right)\bar{\omega}_e x_e - \left(1+\frac{1}{2}\right)\bar{\omega}_e + \left(1+\frac{1}{2}\right)\bar{\omega}_e x_e$$

$$= \bar{\omega}_e (1-4x_e) \quad cm^{-1}.$$

2018, 2021, 2019(t)

(*) What are P, Q, R branches in the vibration rotation spectra? Explain their origin?

⇒ P branch: Lines to the lower frequency side of band origin ($\bar{\omega}_0$), corresponding to negative m (that is $\Delta J = -1$) are referred to P branch. The P branch lines appear at lower frequencies than the pure vibrational transition. Appear at lower frequencies than pure vibrational transition.

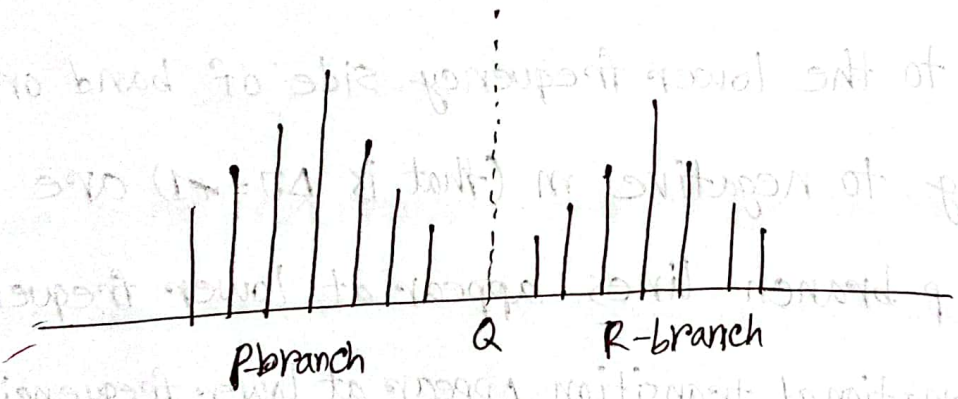
Q branch: In Q branch, rotational quantum number remains unchanged ($\Delta J=0$) during vibrational transition. Q branch is only observed for linear & spherical top molecules. This results in a series of closely spaced lines near the pure vibrational transition frequency.

R branch: Lines to the higher frequency side of band

origin ($\bar{\omega}_0$), corresponding to positive m (that is $\Delta J = +1$) are referred to as R branch. In R branch, J increases by one unit ($\Delta J = +1$) during vibrational transition. this means molecules gains both vibrational & rotational energy. the R branch appear at higher frequencies than the pure Vibrational transition.

Lines arising from $\Delta J = -2, -1, 0, +1, +2$

called O P Q R S branch



2017, 2020

Ⓧ Explain why C=O stretching vibration of Aldehyde gives a strong absorption in IR, although absorption due to C=C vibration in alkene is normally very weak?

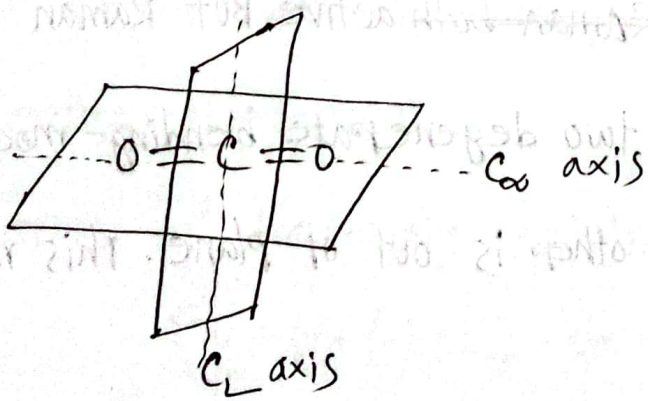
⇒ C=O bond has significant polarity due to electronegativity difference btwn C & O. This results large dipole moment. when the C=O bond stretches, the dipole moment will be changes. The large change in dipole moment during vibration leads to a strong

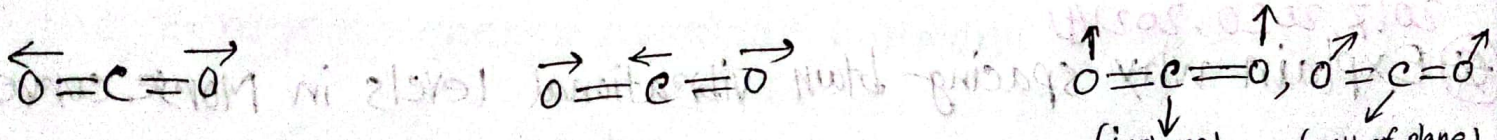
absorption of IR radiation.

the C=C bond in an alkene is not as polar as C=O bond bond since the two C atoms cary almost identical changes; exactly identical if the substituents are same on both C. So ~~when~~ C=C has smaller dipole moment or in some cases almos no dipole moment so when the C=C bond stretches, the changes in dipole moment is very small. This is because the bond length changes but the e^- distribution remains relatively symmetric. Smaller changes in dipole moment during vibration result in a weaker absorption of IR radiation.

~~2017, 2020~~ Sketch symmetry & fundamental vibrations of CO_2 in a figure? explain them clearly?

⇛





(a) symmetric stretch (b) antisymmetric stretch (c) Bending

Fig: The symmetry & fundamental vibrations of CO_2.

total fundamental vibration are = $3N-5$ [as linear]

$= 3\times3-5$

$= 4$

No. of stretching vibration = $N-1 = 3-1 = 2$

No. of Bending mode vibration = $2N-4 = 2$ [as linear]

For CO_2 there are two different sets of symmetry axes. There is an infinite number of two fold axes (C_2) passing through the C atom at right angles to the bond direction, & there is an ∞-fold axis (C_∞) passing through the bond axis itself.

In symmetric stretch Both 'O' atom move toward & away from the C atom simultaneously. This mode is IR inactive but Raman active.

In assymmetric stretch one O moves toward C & other O moves away from C. This mode is IR ~~RAMAN~~ active. But Raman inactive.

Bending mode : There are two degenerate bending modes. one in the plane of the paper & other is out of plane. This mode is both IR & RAman active.

2017, 2020, 2021(T)

(X) Explain why spacing btwn vibrational levels in Morse curve decreases gradually? What are zero point energies in relation to harmonic & anharmonic oscillators?

⇒ The morse curve represents how molecules vibrate. At lower energies, the vibrational levels are evenly apart, as the molecule behaves as a perfect spring. But as the molecule vibrates more, & more, the bonds btwn the atoms stretch & does not behave like a perfect spring anymore. The further it stretches, the less energy it is needed to make it vibrate. Hence the energy difference btwn the vibrational levels get smaller as we go up to higher levels, meaning that the levels are packed more closely together. so the vibrations get stronger, the bond length increases due to stretching & so energy levels get closer. Due to potential energy of a real diatomic molecule is anharmonic. this means that as the vibrational energy increases, the vibrational energy spacing decreases. + Graph.

⇒ energy of harmonic oscillator is -

$$\varepsilon_v = \left(v + \frac{1}{2}\right) \bar{\omega}_{osc} \ cm^{-1} \quad (v = 0, 1, 2, \ldots)$$

So the zero point energy ($v=0$) for harmonic oscillator,

$$\varepsilon_0 = \left(0 + \frac{1}{2}\right) \bar{\omega}_{osc}$$

$$\therefore \varepsilon_0 = \frac{1}{2} \bar{\omega}_{osc}.$$

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energy for anharmonic oscillator

$$\varepsilon_v = \left(v+\frac{1}{2}\right)\bar{\omega}_e - \left(v+\frac{1}{2}\right)^2 \bar{\omega}_e x_e \ \text{cm}^{-1} \quad (v=0,1,2,\cdots)$$

∴ zero point energy ($v=0$) for anharmonic oscillator,

$$\varepsilon_0 = \frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e$$

⊛ 2022, 2015 Write & compare energy of harmonic & anharmonic oscillator. Using Morse curve, Mention the allowed vibrational energy levels & some transitions in anharmonic. write the name of band observed by this transition?

⇒ energy for harmonic, $\varepsilon_v = \left(v+\frac{1}{2}\right)\bar{\omega}_{osc.}\ \text{cm}^{-1}$

energy for anharmonic, $\varepsilon_v = \left(v+\frac{1}{2}\right)\omega_e - \left(v+\frac{1}{2}\right)^2 \bar{\omega}_e x_e \ \text{cm}^{-1}$

compare the energy of harmonic & anharmonic:

$$\bar{\omega}_{osc.} = \bar{\omega}_e\left\{1 - x_e\left(v+\tfrac{1}{2}\right)\right\}$$

Thus anharmonic oscillator behaves like the harmonic oscillator but with an oscillation frequency which decreases steadily with increasing v. Harmonic oscillator have equally spaced energy levels, while anharmonic oscillators have unequally spaced energy levels.

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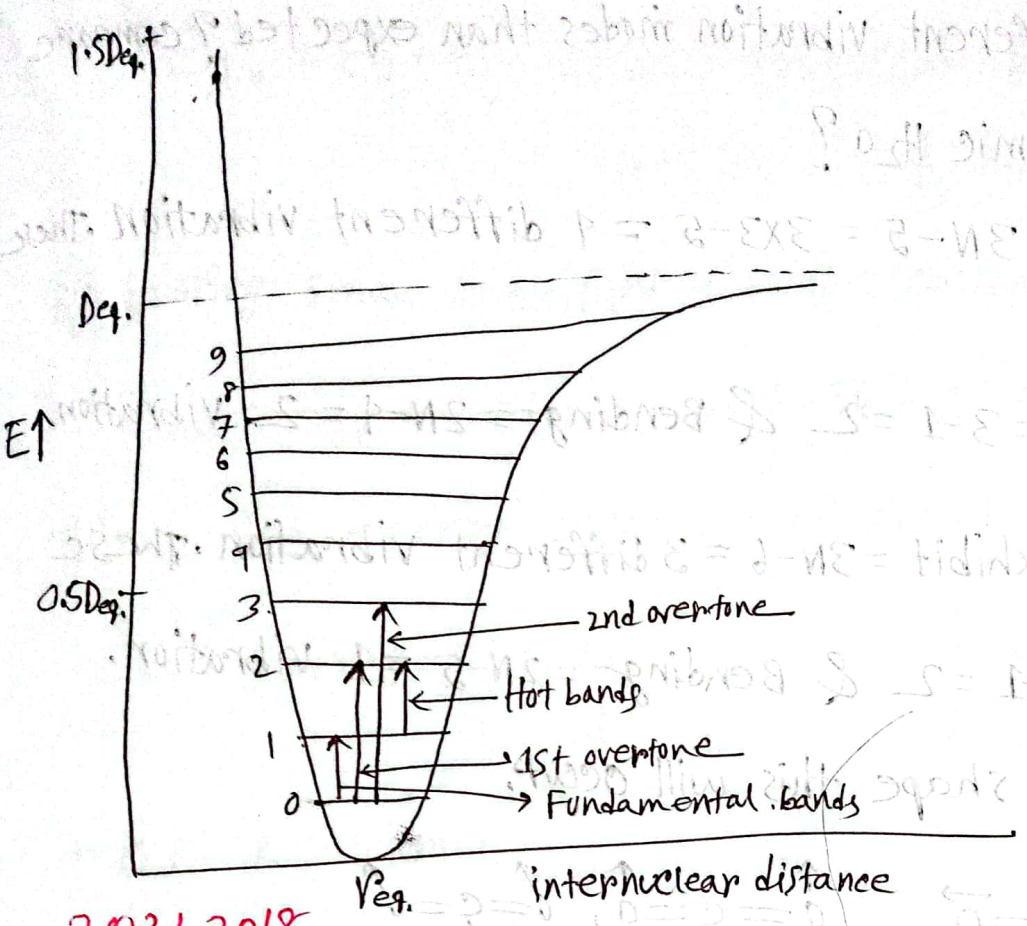


2021, 2018

(X) How can inter & intramolecular H-bonding indentified by IR?

⇒ Intermolecular H-bonding produces broad bands, while intramolecular H-bonding produces sharp bonding.

Inter & intra-molecular H-bonding can also be identified by dilution. Intra-molecular H-bonding remains unaffected by dilution & so the absorption band also remains unaffected. whereas ~~##~~ intermolecular H-bonds are broken on dilution & so there is a decrease in the bonded O-H absorption.

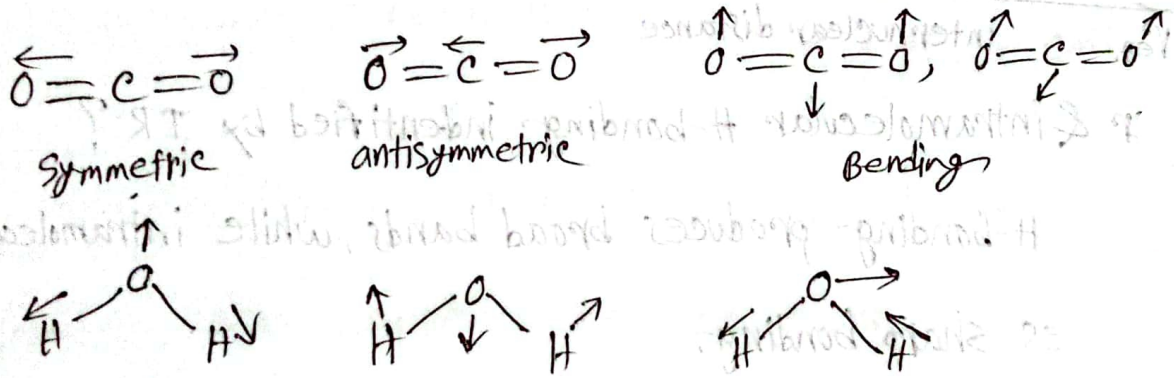
2022
Why CO_2 exhibit different vibration modes than expected? compare with the similar triatomic H_2O?

⇒ Linear CO_2 exhibit $= 3N-5 = 3\times3-5 = 4$ different vibration. These

Stretching $= N-1 = 3-1 = 2$ & Bending $= 2N-4 = 2$ vibration.

And Non linear H_2O exhibit $= 3N-6 = 3$ different vibration. These are, Stretching $= N-1 = 2$ & Bending $= 2N-5 = 1$ vibration.

Due to difference in shape this will occur.



2022
write the energy of a diatomic vibrating rotator? How do you observe the O,P,Q,R,S bands for this molecule?

⇒ A diatomic molecule can execute rotations & vibrations independently. The combined vibrational-rotational energy in the sum of the separate energies, which is:

$$E_{total} = E_{vib.} + E_{rot.} \quad J$$

$$\varepsilon_{total} = \varepsilon_{vib.} + \varepsilon_{rot.} \quad cm^{-1}$$

Taking the separate expressions for $\varepsilon_{rot.}$ & $\varepsilon_{vib.}$, we get,

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$$\varepsilon_{J,v} = \varepsilon_J + \varepsilon_v = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + \ldots\ldots$$
$$+\left(v+\tfrac{1}{2}\right)\bar{\omega}_e - \left(v+\tfrac{1}{2}\right)^2 \bar{\omega}_e x_e \quad cm^{-1}$$

By ignoring small centrifugal distortion constant D, H, etc. Hence we write,

$$\varepsilon_{total} = \varepsilon_{J,v} = BJ(J+1) + \left(v+\tfrac{1}{2}\right)\bar{\omega}_e - \left(v+\tfrac{1}{2}\right)^2 \bar{\omega}_e x_e$$

this is the energy expression of diatomic vibrating rotator.

Lines arising from $\Delta J = -2, -1, 0, +1, +2$
called : O P Q R S branch

+ O, P, Q, R, S describe.

2016, 2019(t), 2018(t)
⊛ State the basic requirement for substance to be IR active with example?

⇒ For IR active, it must ~~have~~ exhibit a change in the dipole moment during molecular vibrations. Vibration that do not result in a dipole moment change are IR active. Ex: HCl is polar with permanent dipole moment. During vibration bond length changes, which alters the dipole moment. This makes HCl IR-active. CO_2's asymmetric stretch resuts change in dipole moment making this mode is IR-active. But Symmetric stretch of CO_2 does not change dipole moment so this mode of vibration is IR inactive.

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Extra

⊗ The fundamental mode of HCl occurs at 2886 cm^{-1}. Predict the frequency of the corresponding mode of DCl?

⇒ Let, K is equal for H-Cl & D-Cl

$$\frac{\nu_{H-Cl}}{\nu_{D-Cl}} = \frac{\sqrt{\frac{K_1}{\mu_1}}}{\sqrt{\frac{K_2}{\mu_2}}} = \sqrt{\frac{\mu_2}{\mu_1}} \quad \text{—— (1)}$$

$$\mu_1 = \frac{1\times35}{1+35} = \frac{35}{36}, \quad \mu_2 = \frac{2\times35}{2+35} = \frac{70}{37}$$

$$\therefore \frac{\nu_{H-Cl}}{\nu_{D-Cl}} = \sqrt{\frac{\frac{70}{37}}{\frac{35}{36}}} = \sqrt{\frac{70\times36}{37\times35}} = 1.39$$

$$\Rightarrow \nu_{D-Cl} = \frac{\nu_{H-Cl}}{1.39} = \frac{2886}{1.39} = 2068\,cm^{-1} \text{ Ans.}$$

2016

⊗ Explain why all transitions in vibrational spectroscopy are restricted to originate from ground state (V=0)?

⇒ because at room temperature, only the lowest energy vibrational stat is populated. Due to population distribution & selection rules. The combination of Boltzmann distribution & selection rule results in the dominance of transitions from ground state (V=0) to the 1st excited state. Higher transitions such as V=0 to V=2 or V=1 to V=2 are less probable due to lower population of higher vibrational levels. while hot bands can be observed at higher temperature. they are significantly weaker base: The intensity of these transitions decreases accordingly. It occur due to the low population of higher vibrational states at standard temperatures & large gap btwn vibrational levels.

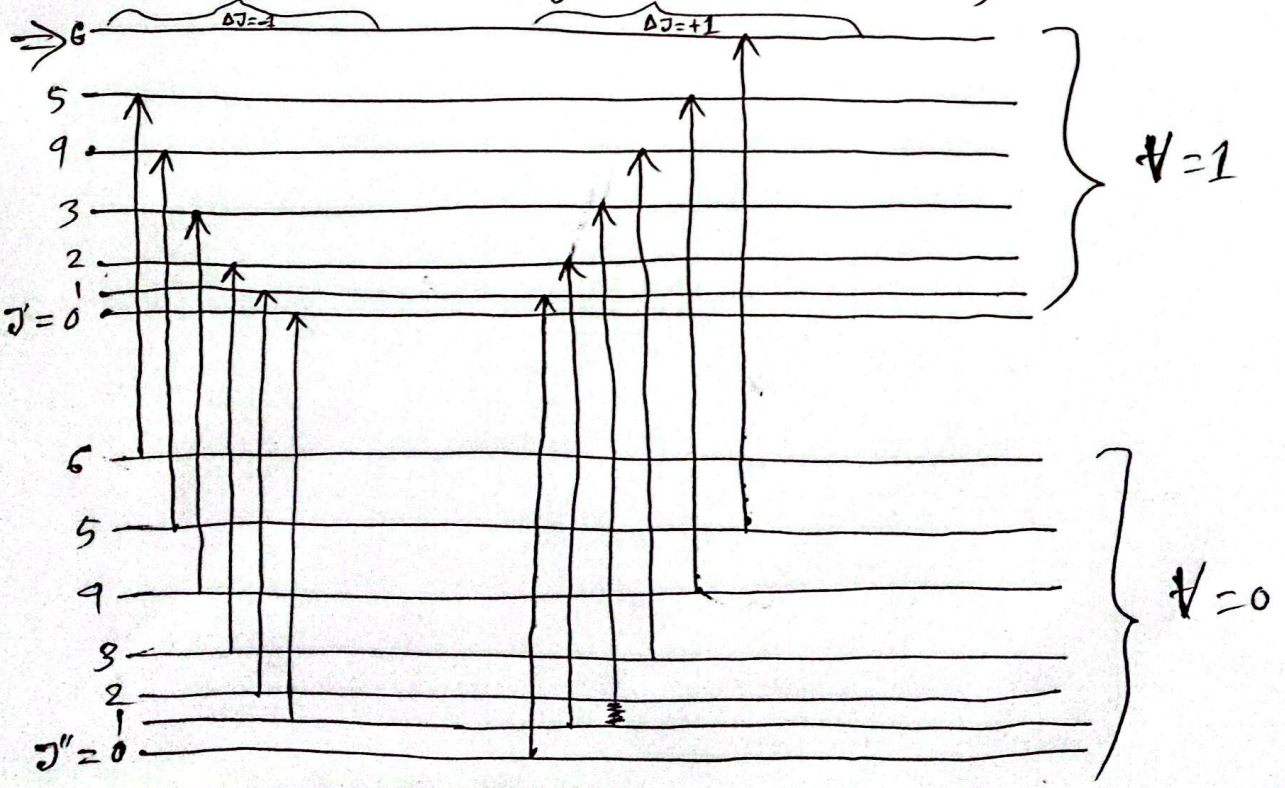
2016

(8) what are P & R branch lines? Sketch vibrational-rotational energy levels & show transitions btwn them to appear P & R branch lines in vibration-rotation spectrum?

⇒ P & R branch lines refer to specific types of spectral lines observed in molecular spectroscopy, particularly in rotational-vibrational spectroscopy of diatomic molecules.

P branch corresponds to transitions where the rotational quantum number decreases by 1 ($\Delta J = -1$). Transition occur from higher rotational state to lower state ($J \rightarrow J-1$).

R branch corresponds to transitions where the rotational quantum number increases by 1 ($\Delta J = +1$). Transition occur from a lower rotational state to a higher one ($J \rightarrow J+1$).



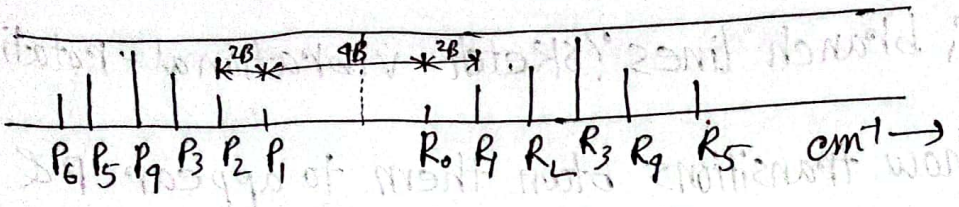


Fig: some transitions btwn the rotational vibrational energy levels of a diatomic molecule, together with the spectrum arising from them.

2018(t)

A soln gives absorbance 0.459 at 359 nm in a 1 cm cell. the molar absorptivity is 1.25×10^4 cm^{-1} $mol^{-1}L$. what is concentration?

$$\Rightarrow A = \varepsilon c l \Rightarrow c = \frac{A}{\varepsilon l} = \frac{0.459}{1.25 \times 10^4 \times 1} = 3.632 \times 10^{-5} M$$

Ans.